

Graph Algorithm and Applications

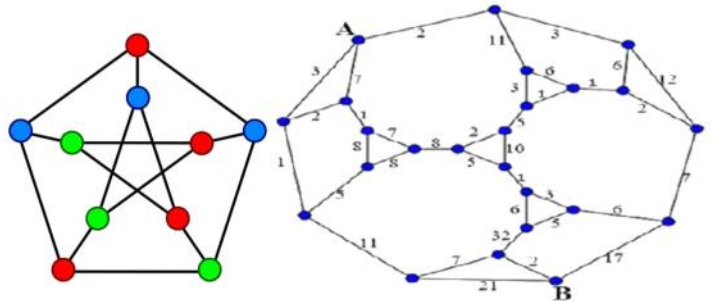
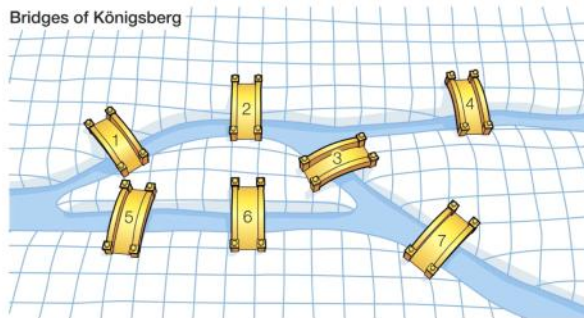
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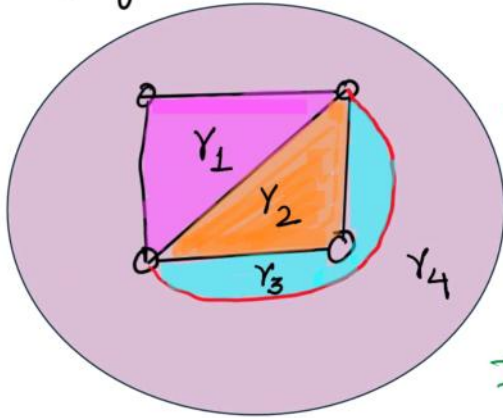
Bridges of Königsberg



Graph Theory: The Language of Connections

Region of Plane Graphs

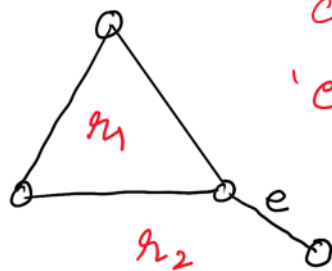
They are the sections in which a planar graph is divided. OR, it is a subgraph with vertices and edges that are incident with the region.



Regions are also called as faces of a plane graph. R_4 - unbounded exterior region.

NOTE:- All the edges are part of boundaries of 2 regions.

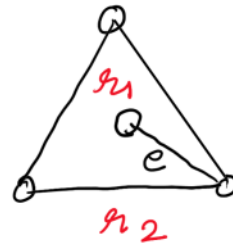
Is this always true?



count the no. of regions : 2
'e' is part of : r_2

Now make an isomorphic graph

Now 'e' is part of : r_1



Point:- Isomorphism sometimes changes the region of edges.

Observation

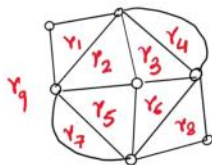
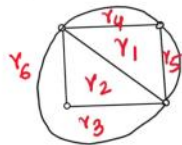
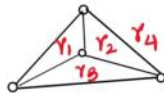
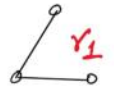
F - face or region

For a plane graph

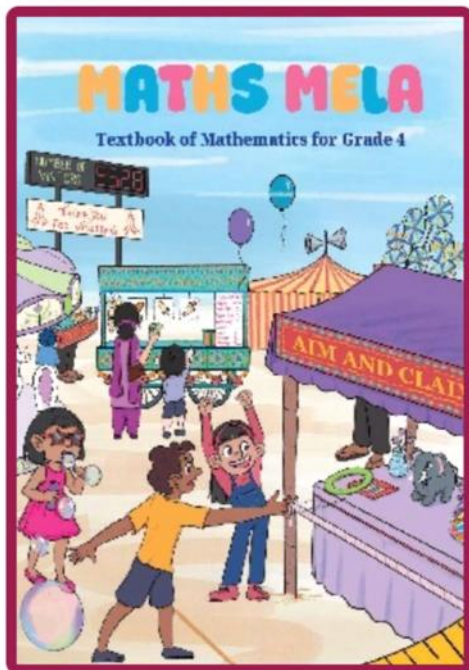
$$V - E + F = 2$$

Q:- Have you seen this formula earlier?

Yes



V	E	F	$V - E + F$
2	1	1	$2 - 1 + 1 = 2$
3	2	1	$3 - 2 + 1 = 2$
3	3	2	$3 - 3 + 2 = 2$
4	6	4	$4 - 6 + 4 = 2$
4	8	6	$4 - 8 + 6 = 2$
7	14	9	$7 - 14 + 9 = 2$



EULER'S FORMULA

$$\text{FACES} + \text{VERTICES} - \text{EDGES} = 2$$



TETRAHEDRON

$$4 + 4 - 6 = 2$$



CUBE

$$6 + 8 - 12 = 2$$



OCTAHEDRON

$$8 + 6 - 12 = 2$$



DODECAHEDRON

$$12 + 20 - 30 = 2$$



ICOSAHEDRON

$$20 + 12 - 30 = 2$$

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Euler's Formula for Plane Graphs

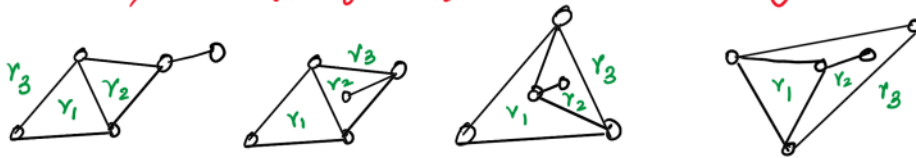
Let G be a plane drawing (or plane graph) of a connected graph, and let n , m , and f (or r) be the number of vertices, edges, and regions then:

$$n - m + f(\text{or } r) = 2$$

This formula also tells that :-

For any planar connected graph G {
 n - fixed
 m - fixed

Hence, r (or f) gets fixed (or unchanged)



Euler's Formula Proof

The method adopted is called minimum counter example.

↓
Proof by contradiction

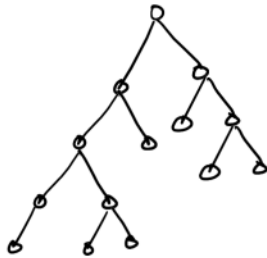
The second popular method is called as proof by induction.

In both the methods, one edge has to be removed to make a connected graph with fewer edges.

First we will be proving it for a tree graph.

Let G be a tree with

- n Vertices
- Edges $= n-1$
- regions $= 1$



$$\therefore n - m + f = n - (n-1) + 1$$

$$= n - n + 2$$

$$\text{Hence, } n - m + f(\text{or } r) = 2$$

$\therefore$ Tree is a planar graph.

For graphs that are not trees

Let say that $m - n + f \text{ (or } r) \neq 2$ for some graph (G) which is connected, having n vertices, f regions, and a min. size m .

Let G has a cycle with an edge ' e '. → Is this True yes

Now consider a graph $G - e$

Vertices	n
Edges	$m - 1$
faces/regions	$f - 1$

As $G - e$ is a connected plane graph and its size is less than m which means that it will satisfy the Euler's formula.

Removal of 1 edge reduces the faces by 1

$$\therefore n - (m - 1) + f - 1 = 2$$

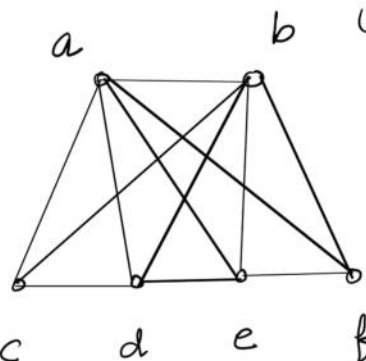
$$\leftarrow n - m + 1 + f - 1 = 2$$

$$n - m + f = 2$$

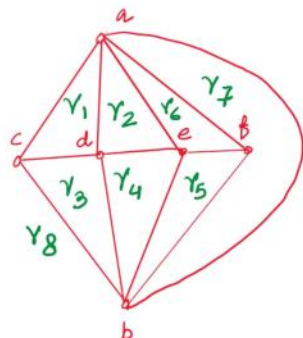
Hence, there is no minimum counter example and thus, there can be no counter examples.

Verify Eulers formula for the following graphs:

(1)

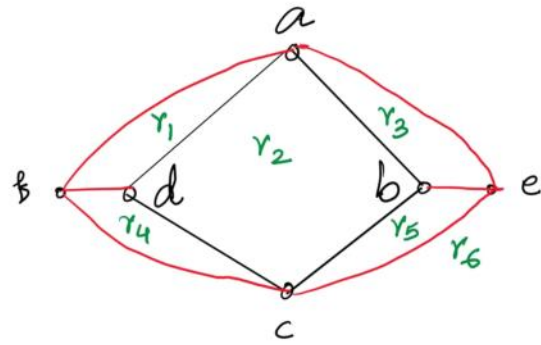
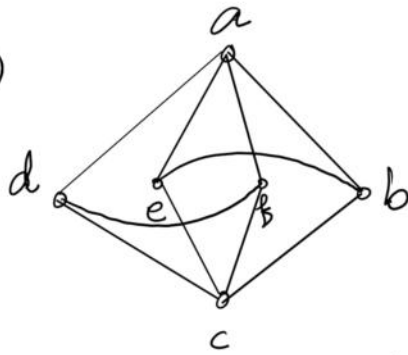


(i) Draw the plane graph for this graph.



(ii) count the
 Vertices $\rightarrow 6$
 Edges $\rightarrow 12$
 faces $\rightarrow 8$
 $n - m + f = 6 - 12 + 8$
 $\therefore n - m + f = 2$
 verified.

(2)



Vertices (n) = 6

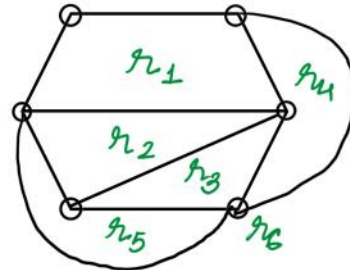
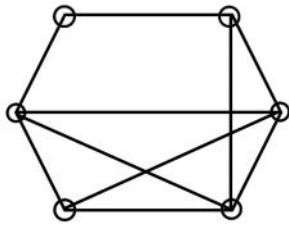
Edges (m) = 10

Regions (f) = 6

$$\therefore n - m + f = 6 - 10 + 6 = 2$$

Verified

3

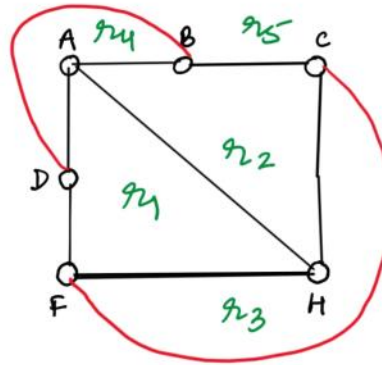
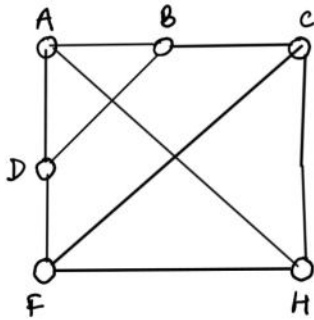


plane graph

Vertices (n) = 6
Edges (m) = 10
faces (f) = 6

$$\therefore n - m + f = 6 - 10 + 6 = 2 \quad \text{verified}$$

④



$$\begin{aligned} n &= 6 \\ m &= 9 \\ f &= 5 \end{aligned}$$

$$\therefore n - m + f = 6 - 9 + 5 = 2$$

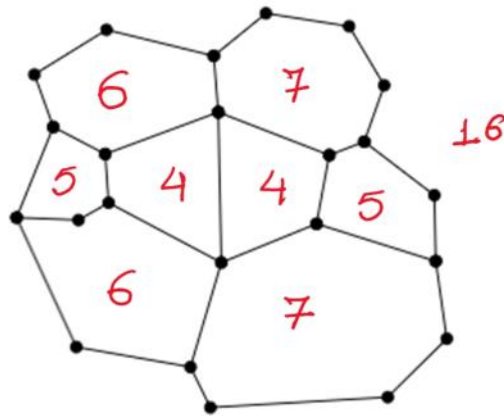
Verified

Degree of a face

The degree of a face f is the number of edges along its boundary.

Alternatively, it is the number of vertices along its boundary.

Alternatively, it is the number of other faces with which it shares an edge.



In all the examples we were able to convert them into plane graphs.

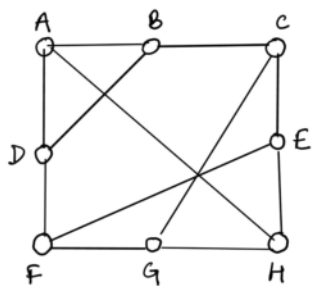
what if the graphs are not planar
or
it is too complicated.

(i.e. we find it difficult to draw plane graph)



Let $G = (V, E)$, be a simple connected planar graph with at least 3 edges, then the following inequality holds:

- $2m \geq 3f$ (or r)
- $m \leq 3n - 6$
- There is always a vertex v in G such that $d(v) \leq 5$



Don't try to convert into plane graph
Thus we do not have value of 'f'

$$n = 8$$

$$m = 12$$

go for 2nd inequality i.e $m < 3n - 6$

RHS :- $3 \times 8 - 6 = 24 - 6 = 18$

$$\therefore 12 < 18 \quad \text{Reason?}$$

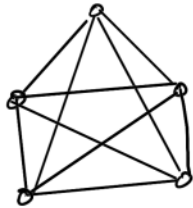
$\therefore$ inequality does holds, But we know that this graph is non-planar.

Euler inequality is a necessary condⁿ for a graph to be planar *but not a sufficient conⁿ.*

Note:- Satisfaction of Euler's inequality does not mean the graph is planar; it only means it might be.

If the inequality fails then it is true that the graph is non-planar.

* Prove that K_5 is a non-planar graph



$$n = 5$$

$$m = \frac{5(5-1)}{2} = 10$$

If Euler's inequality becomes false, then K_5 is NP.
 $m < 3n - 6$ — Euler's IE

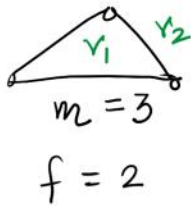
$$3 \times 5 - 6 = 15 - 6 = 9$$

$$\therefore 10 \not< 9$$

Hence, K_5 is NP.

Proof of $m \leq 3n - 6$ (Simple Graph)

Let G be any planar graph having regions $f_1, f_2, \dots, f_r$
 each region will have vertices and edges in its boundaries
 $m_1, m_2, \dots, m_r$



Observation

- (i) Each region is bounded by at least 3 edges. The sum of degrees of the faces is atleast $3 \cdot f$
- (ii) Each edge is shared by at least 2 region.

In general

$$\frac{m}{f} \geq \frac{3}{2}$$

But

$$n - m + f = 2$$

for a planar graph.

$$\text{So, } \frac{m}{f} \geq \frac{3}{2} \rightarrow 2m \geq 3f \rightarrow 2m \geq 3(2-n+m)$$

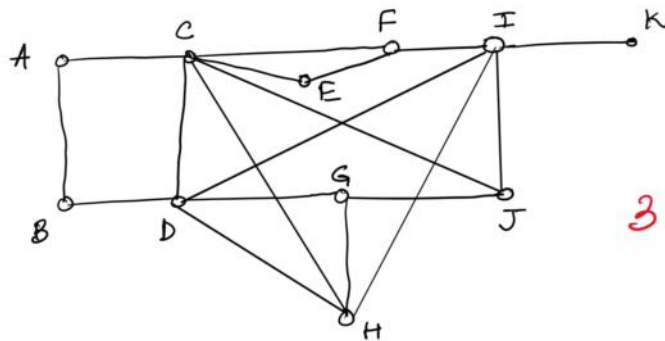
$$\therefore 2m \geq 6 - 3n + 3m$$

$$2m - 3m \geq 6 - 3n$$

$$-m \geq 6 - 3n$$

$$\text{or } \boxed{m \leq -6 + 3n}$$

Check whether the following graphs is planar or not.



$$m = 18$$

$$n = 11$$

$$3n - 6 = -6 + 3 \times 11$$

$$= 6 - 33$$

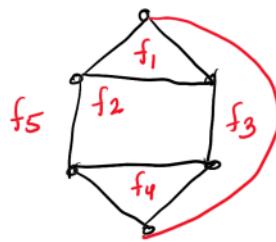
$$= -27$$

$$\text{Hence, } m < -6 + 3n$$

This graph can be planar. But, not certain.

How many regions does a 3-Regular graph with 6 vertices have?

- 3-regular means: degree 3 for each vertex.
- $n = 6$
- $3 \times 6 = 2 \times m \rightarrow m = \frac{18}{2} = 9$
- $m - n + f = 3 \rightarrow 9 - 6 + f = 3 \rightarrow f = 3 - (-3) = 5$
- **ANS.** $f = 5$



A connected cubic (3-regular) planar graph has 12 vertices. Exactly six faces are pentagons; all remaining faces are triangles.

Find: (a) the number of edges, (b) the total number of faces, (c) the number of triangular faces.

Cubic (3-regular) connected planar, $V = 12$. Six pentagonal faces; rest triangles.

Degree sum: $3V = 36 = 2E \Rightarrow E = 18$.

Euler: $V - E + F = 2 \Rightarrow 12 - 18 + F = 2 \Rightarrow F = 8$.

Face handshake: $5 \cdot 6 + 3f_3 = 2E = 36 \Rightarrow 30 + 3f_3 = 36 \Rightarrow f_3 = 2$.

Ans: $E = 18$, $F = 8$, triangular faces = 2.

In a simple connected planar graph, there are 36 edges. Exactly 12 faces are quadrilaterals and every other face is a triangle.

Find: (a) the number of triangular faces, (b) the total number of faces, (c) the number of vertices.

Simple connected planar, $E = 36$. Twelve 4-faces; rest triangles.

Let $f_4 = 12$, $f_3 = x$.

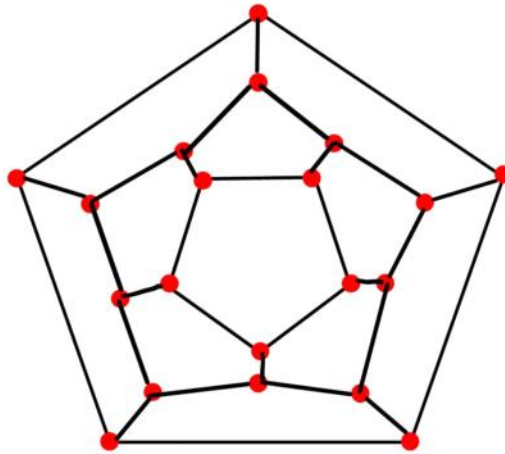
Face handshake: $4 \cdot 12 + 3x = 2E = 72 \Rightarrow 48 + 3x = 72 \Rightarrow x = 8$.

$F = x + 12 = 20$.

Euler: $V - E + F = 2 \Rightarrow V - 36 + 20 = 2 \Rightarrow V = 18$.

Ans: triangles = 8, total faces $F = 20$, vertices $V = 18$.

Verify Eulers equation for the following plane graph. (Homework problem)



A convex polyhedron has 24 vertices. Exactly three faces meet at every vertex, and every face is either a pentagon or a hexagon.

Find: (a) the number of pentagonal faces, (b) the number of hexagonal faces, (c) the total number of edges.

Convex polyhedron, $V = 24$. Exactly three faces meet at each vertex (so $\deg = 3$).

Faces are 5- or 6-gons. Let $p = f_5$, $h = f_6$.

Vertex handshake: $3V = 72 = 2E \Rightarrow E = 36$.

Face handshake: $5p + 6h = 2E = 72$.

Euler: $V - E + F = 2 \Rightarrow 24 - 36 + (p + h) = 2 \Rightarrow p + h = 14$.

Solve: $5p + 6h = 72$, $p + h = 14 \Rightarrow h = 2$, $p = 12$.

Ans: pentagons $p = 12$, hexagons $h = 2$, edges $E = 36$.